Chapter 13

Vectors II

Contents

13.4 Lines . . . 2
13.4.1 Equations of Lines . . . 2
13.4.2 Foot of Perpendicular (Point to Line) . . . 4
13.4.3 Reflection (Point about Line) . . . 6
13.4.4 Shortest Distance (Point to Line) . . . 8
13.4.5 Relationship Between Lines . . . 10
13.4.6 Angle Involving Lines . . . 12
13.4.7 Reflection (Line about Line) . . . 13
13.4.8 Visual Atlas: Lines . . . 15
13.5 Planes . . . 20
13.5.1 Equations of Planes . . . 20
13.5.2 Relationship Between Line and Plane . . . 22
13.5.3 Foot of Perpendicular (Point to Plane) . . . 24
13.5.4 Shortest Distance (Point to Plane) . . . 26
13.5.5 Reflection (Line about Plane) . . . 28
13.5.6 Relationship Between Two Planes . . . 30
13.5.7 Angles Involving Planes . . . 32
13.5.8 Distance Between Parallel Planes . . . 34
13.5.9 Visual Atlas: Planes . . . 36
13.6 Interaction between Three Planes . . . 43

13.4 Lines

13.4.1 Equations of Lines

A line is fixed by **one point on it** and **one direction**. Every equation of a line is a way of writing those two things down.

Definition 13.4.1. Vector form of a line

The vector equation of a line is

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}, \qquad \lambda \in \mathbb{R},$$

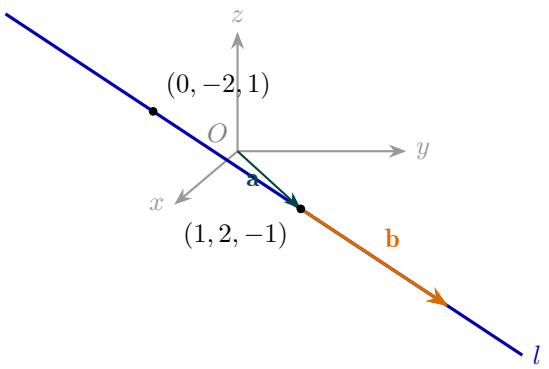
where $\mathbf{a}$ is the position vector of *any* point on the line and $\mathbf{b}$ is the **direction** of the line.

Each value of λ gives one point: $\mathbf{r}$ is the position vector of a general point on the line. Reading the three rows one at a time gives the **parametric form** $x = a_1 + \lambda b_1$, $y = a_2 + \lambda b_2$, $z = a_3 + \lambda b_3$.

Definition 13.4.2. Cartesian form of a line

Make λ the subject of each parametric equation and equate:

$$\frac{x - a_1}{b_1} = \frac{y - a_2}{b_2} = \frac{z - a_3}{b_3} \quad (= \lambda).$$



The equation of a line is not unique. Any point on the line can be $\mathbf{a}$, and any non-zero multiple of $\mathbf{b}$ can be the direction. Two correct answers may look different.
Edge case: a zero component. If $b_1 = 0$, you cannot divide by it. That coordinate is constant along the line: write $x = a_1$ separately, and equate the other two fractions.

Example 13.4.1

Find both the vector and Cartesian equations of the line that passes through

(a) $(1, 2, -1)$ and $(0, -2, 1)$,

(b) the origin, and is parallel to the vector $3\mathbf{i} + \mathbf{j} - \mathbf{k}$,

(c) $(1, 0, 1)$, and is perpendicular to both $-\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $4\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

Solution. (a) Direction $\begin{pmatrix}0\\-2\\1\end{pmatrix} - \begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}-1\\-4\\2\end{pmatrix}$, or more simply $\begin{pmatrix}1\\4\\-2\end{pmatrix}$.

$$\mathbf{r} = \begin{pmatrix}1\\2\\-1\end{pmatrix} + \lambda\begin{pmatrix}1\\4\\-2\end{pmatrix},\ \lambda \in \mathbb{R}; \qquad x-1 = \frac{y-2}{4} = \frac{z+1}{-2}.$$

(b) The origin is on the line, so $\mathbf{a} = \mathbf{0}$:

$$\mathbf{r} = \mu\begin{pmatrix}3\\1\\-1\end{pmatrix},\ \mu \in \mathbb{R}; \qquad \frac{x}{3} = y = -z.$$

(c) A direction perpendicular to both vectors is their vector product:

$$\begin{pmatrix}-1\\1\\2\end{pmatrix} \times \begin{pmatrix}4\\1\\2\end{pmatrix} = \begin{pmatrix}2-2\\-(-2-8)\\-1-4\end{pmatrix} = \begin{pmatrix}0\\10\\-5\end{pmatrix} = 5\begin{pmatrix}0\\2\\-1\end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix}1\\0\\1\end{pmatrix} + \alpha\begin{pmatrix}0\\2\\-1\end{pmatrix},\ \alpha \in \mathbb{R}; \qquad \frac{y}{2} = \frac{z-1}{-1},\quad x = 1.$$

Common mistake: losing the sign of a negative component in the Cartesian form. For $\mathbf{r} = \mu(3\mathbf{i} + \mathbf{j} - \mathbf{k})$:

$$\frac{x}{3} = \frac{y}{1} = \frac{z}{1},\ \text{i.e.}\ \frac{x}{3} = y = z \qquad \times$$
$$\frac{x}{3} = \frac{y}{1} = \frac{z}{-1},\ \text{i.e.}\ \frac{x}{3} = y = -z \qquad \checkmark$$

Check one point: $\mu = 1$ gives $(3, 1, -1)$, and $-z = 1 = y$.

Example 13.4.2

Determine whether the points $P(3, 10, -5)$ and $Q(2, 5, -3)$ lie on the line $\mathbf{r} = \begin{pmatrix}1\\2\\-1\end{pmatrix} + \lambda\begin{pmatrix}1\\4\\-2\end{pmatrix}$.

Solution. Solve for λ in each row; the point is on the line only if **all three** give the same λ.

P: $1 + \lambda = 3$, $2 + 4\lambda = 10$, $-1 - 2\lambda = -5$ all give $\lambda = 2$. So P lies on the line.

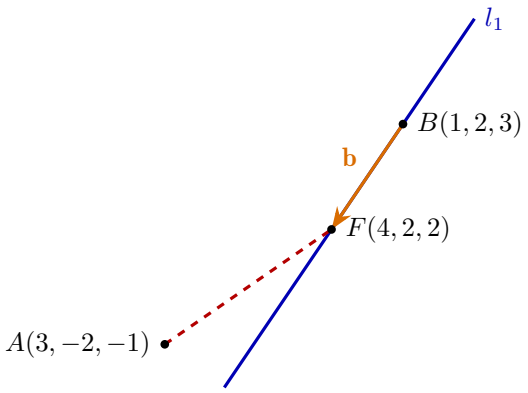
Q: $1 + \lambda = 2$ gives $\lambda = 1$, but $2 + 4\lambda = 5$ gives $\lambda = \frac{3}{4}$. So Q does not lie on the line.

13.4.2 Foot of Perpendicular (Point to Line)

Given a point A and a line l, the **foot of the perpendicular** from A to l is the point F on l with $AF \perp l$. It is the point of l closest to A.

Definition 13.4.3. Method: foot of perpendicular from A to l : $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$

1. F lies on l, so $\overrightarrow{OF} = \mathbf{a} + \lambda\mathbf{b}$ for some value of λ.
2. Write $\overrightarrow{AF} = \overrightarrow{OF} - \overrightarrow{OA}$, in terms of λ.
3. $\overrightarrow{AF} \perp l$, so $\overrightarrow{AF} \cdot \mathbf{b} = 0$. Solve for λ.
4. Substitute λ back into $\overrightarrow{OF}$.



The picture is Example 13.4.3: the dashed segment AF meets l_1 at a right angle (in space; a flat drawing cannot show every right angle as one).

Example 13.4.3

Find the foot of perpendicular of the point $A(3,-2,-1)$ on the line

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \ \lambda \in \mathbb{R}.$$

Solution. Let F be the foot. Since F lies on l_1, $\overrightarrow{OF} = \begin{pmatrix} 1+3\lambda \\ 2 \\ 3-\lambda \end{pmatrix}$ for some λ, and

$$\overrightarrow{AF} = \begin{pmatrix} 1+3\lambda \\ 2 \\ 3-\lambda \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3\lambda - 2 \\ 4 \\ 4-\lambda \end{pmatrix}.$$

$\overrightarrow{AF} \perp l_1$: $\begin{pmatrix} 3\lambda-2 \\ 4 \\ 4-\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = 9\lambda - 6 - 4 + \lambda = 10\lambda - 10 = 0$, so $\lambda = 1$.

$\overrightarrow{OF} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$: the foot is $\begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$, i.e. $F(4,2,2)$.

Example 13.4.4

Find the coordinates of the foot of perpendicular of the point $P(-2,1,1)$ on the line

$$l_1: \mathbf{r} = \begin{pmatrix} -4 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \ \mu \in \mathbb{R}.$$

Solution. $\overrightarrow{OF} = \begin{pmatrix} -4 \\ 1+\mu \\ -\mu \end{pmatrix}$, so $\overrightarrow{PF} = \begin{pmatrix} -4 \\ 1+\mu \\ -\mu \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ \mu \\ -\mu - 1 \end{pmatrix}$.

$\overrightarrow{PF} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 0 + \mu + \mu + 1 = 2\mu + 1 = 0$, so $\mu = -\frac{1}{2}$.

$\overrightarrow{OF} = \begin{pmatrix} -4 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$: the foot is $\begin{pmatrix} -4 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$, i.e. $F\left(-4, \frac{1}{2}, \frac{1}{2}\right)$.

Common mistake: dotting the *position vector* of F with the direction, instead of $\overrightarrow{AF}$. In Example 13.4.3:

$$\overrightarrow{OF} \cdot \mathbf{b} = 3(1+3\lambda) - (3-\lambda) = 10\lambda = 0 \ \Rightarrow \ \lambda = 0, \ F(1,2,3) \quad \times$$
$$\overrightarrow{AF} \cdot \mathbf{b} = 10\lambda - 10 = 0 \ \Rightarrow \ \lambda = 1, \ F(4,2,2) \quad \checkmark$$

$\overrightarrow{OF} \cdot \mathbf{b} = 0$ finds the foot of the perpendicular from the *origin*, not from A.

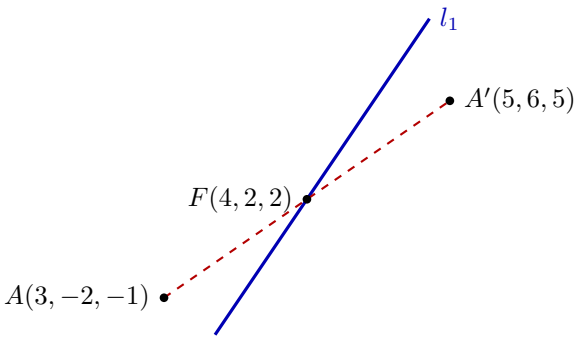
13.4.3 Reflection (Point about Line)

To reflect a point A about a line, first find the foot F (Section 13.4.2). The image A' is on the other side of the line, as far from F as A is, so F is the **midpoint** of AA'.

Definition 13.4.4. Reflection of a point about a line

If F is the foot of the perpendicular from A to the line, and A' is the reflection of A, then

$$\overrightarrow{OF} = \frac{\overrightarrow{OA} + \overrightarrow{OA'}}{2} \quad \Longrightarrow \quad \overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA}.$$



Example 13.4.5

Find the reflection of the point $A(3,-2,-1)$ about the line $l_1: \mathbf{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}3\\0\\-1\end{pmatrix}$, $\lambda \in \mathbb{R}$.

Solution. From Example 13.4.3, the foot is $F(4,2,2)$. Then

$$\begin{aligned}\overrightarrow{OF} &= \frac{\overrightarrow{OA} + \overrightarrow{OA'}}{2}\\ \overrightarrow{OA'} &= 2\overrightarrow{OF} - \overrightarrow{OA}\\ &= 2\begin{pmatrix}4\\2\\2\end{pmatrix} - \begin{pmatrix}3\\-2\\-1\end{pmatrix}\\ &= \begin{pmatrix}5\\6\\5\end{pmatrix}\end{aligned}$$

so $A'(5,6,5)$. Check: $\frac{1}{2}\big((3,-2,-1)+(5,6,5)\big) = (4,2,2) = F$.

Example 13.4.6

Find the reflection of the point $P(-2,1,1)$ about the line $l_1: \mathbf{r} = \begin{pmatrix}-4\\1\\0\end{pmatrix} + \mu\begin{pmatrix}0\\1\\-1\end{pmatrix}$, $\mu \in \mathbb{R}$.

Solution. From Example 13.4.4, the foot is $F\left(-4, \frac{1}{2}, \frac{1}{2}\right)$. Then

$$\begin{aligned}\overrightarrow{OF} &= \frac{\overrightarrow{OP} + \overrightarrow{OP'}}{2} \\ \overrightarrow{OP'} &= 2\overrightarrow{OF} - \overrightarrow{OP} \\ &= \begin{pmatrix} -8 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -6 \\ 0 \\ 0 \end{pmatrix}\end{aligned}$$

so $P'(-6, 0, 0)$.

Common mistake: subtracting the wrong way round, $\overrightarrow{OA'} = \overrightarrow{OA} - 2\overrightarrow{OF}$. In Example 13.4.6:

$$\overrightarrow{OP'} = \overrightarrow{OP} - 2\overrightarrow{OF} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -8 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} \quad \times$$

$$\overrightarrow{OP'} = 2\overrightarrow{OF} - \overrightarrow{OP} = \begin{pmatrix} -8 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ 0 \\ 0 \end{pmatrix} \quad \checkmark$$

Check the midpoint: $\frac{1}{2}\big((-2, 1, 1) + (6, 0, 0)\big) = \left(2, \frac{1}{2}, \frac{1}{2}\right)$, which is not F. Always check that F is the midpoint.

Example 13.4.7

Find the reflection of the origin O about the line $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$.

Solution. Here the point *is* O, so $\overrightarrow{OF} \cdot \mathbf{b} = 0$ is the right equation: $3(1 + 3\lambda) - (3 - \lambda) = 10\lambda = 0$, so $\lambda = 0$ and $F(1, 2, 3)$. Then $\overrightarrow{OO'} = 2\overrightarrow{OF} - \mathbf{0} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$, so $O'(2, 4, 6)$.

Intermediate step. Every reflection question is a foot-of-perpendicular question first. The same midpoint formula reflects a point about a plane (Section 13.5.3); only the way of finding F changes.

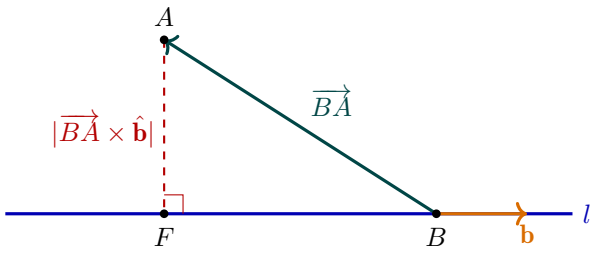
13.4.4 Shortest Distance (Point to Line)

The shortest distance from a point A to a line is the perpendicular distance AF. Section 13.3.2 gives it without finding F: take *any* point B on the line, and use the vector product.

Definition 13.4.5. Shortest distance from A to l : $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$

Let B be any point on l (for example the point with position vector $\mathbf{a}$). Then

$$\text{Shortest distance} = \left|\overrightarrow{BA} \times \hat{\mathbf{b}}\right| = \frac{\left|\overrightarrow{BA} \times \mathbf{b}\right|}{|\mathbf{b}|}.$$



Two routes. If the foot F is already known (say from an earlier part), the distance is simply $|\overrightarrow{AF}|$. If it is not, finding F just to get a distance is unnecessary and tedious: use the cross product.

Example 13.4.8

Find the shortest distance from the point $A(3, -2, -1)$ to the line l_1 : $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$, $\lambda \in \mathbb{R}$.

Solution. Take $B(1, 2, 3)$ on l_1. $\overrightarrow{AB} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix}$.

$$\text{Distance} = \frac{\left|\begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix} \times \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}\right|}{\left|\begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} -4 \\ 10 \\ -12 \end{pmatrix}\right|}{\sqrt{10}} = \frac{\sqrt{260}}{\sqrt{10}} = \sqrt{26} \text{ units.}$$

($\overrightarrow{AB}$ and $\overrightarrow{BA}$ give the same modulus.) Check with the foot $F(4, 2, 2)$: $|\overrightarrow{AF}| = \left|\begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}\right| = \sqrt{26}$.

Example 13.4.9

Find the perpendicular distance from the point $P(-2,1,1)$ to the line $l_1:\ \mathbf{r} = \begin{pmatrix} -4 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix},\ \mu \in \mathbb{R}$.

Solution. Take $B(-4,1,0)$ on l_1. $\overrightarrow{BP} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -4 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$.

$$\text{Length} = \frac{\left| \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right|}{\left| \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} \right|}{\sqrt{2}} = \frac{3}{\sqrt{2}} \text{ units.}$$

Common mistake: dividing by the wrong length, $|\overrightarrow{BP}|$ instead of $|\mathbf{b}|$.

$$\text{Distance} = \frac{|\overrightarrow{BP} \times \mathbf{b}|}{|\overrightarrow{BP}|} = \frac{3}{\sqrt{5}} \qquad \times$$

$$\text{Distance} = \frac{|\overrightarrow{BP} \times \mathbf{b}|}{|\mathbf{b}|} = \frac{3}{\sqrt{2}} \qquad \checkmark$$

It is the *line's direction* that becomes a unit vector: $|\overrightarrow{BP} \times \hat{\mathbf{b}}|$.

13.4.5 Relationship Between Lines

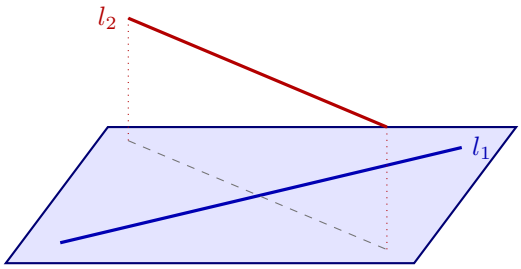
In a plane, two lines either meet or are parallel. In space there is a third possibility.

Definition 13.4.6. Two lines in space

Two lines l_1 and l_2 are either

1. **parallel** (their directions are parallel)*,
2. **intersecting** (most common), or
3. **skew**: not parallel, and they do not intersect.

*Parallel lines may be two distinct lines, or exactly the same line.



Skew lines: l_2 passes above l_1. Seen from above (dashed shadow) they seem to cross, but they never meet.

Definition 13.4.7. How to tell

- Check if the directions of the lines are parallel. If so, the lines are case (1)*. Then check whether a point of one line lies on the other: if it does, they are the same line.
- If not, they are either (2) or (3). Equate the general points, $\mathbf{a}_1 + \lambda\mathbf{b}_1 = \mathbf{a}_2 + \mu\mathbf{b}_2$, solve two of the three equations for λ and μ, and **check the third**. Consistent: they intersect. Inconsistent: they are skew.

Example 13.4.10

Find the intersection between l_1 and l_2, where

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \ \lambda \in \mathbb{R}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 12 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}, \ \mu \in \mathbb{R}.$$

Solution. The directions are not parallel. At an intersection,

$$\begin{pmatrix} 1+\lambda \\ 2\lambda \\ 2+\lambda \end{pmatrix} = \begin{pmatrix} 2-\mu \\ 12+3\mu \\ 5 \end{pmatrix} \implies \begin{cases} \lambda + \mu = 1 \\ 2\lambda - 3\mu = 12 \\ \lambda = 3 \end{cases}$$

The third equation gives $\lambda = 3$, then the first gives $\mu = -2$. Check the second: $2(3) - 3(-2) = 12$. ✓

The lines intersect at $\begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix}$, i.e. $(4, 6, 5)$.

Example 13.4.11

For $l_1 :\ \mathbf{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda \begin{pmatrix}1\\2\\1\end{pmatrix}$, describe the relationship between l_1 and

(a) $l_3 :\ \mathbf{r} = \begin{pmatrix}2\\12\\7\end{pmatrix} + \mu \begin{pmatrix}-1\\3\\0\end{pmatrix}$,

(b) $l_4 :\ \mathbf{r} = t \begin{pmatrix}2\\4\\2\end{pmatrix}$,

(c) $l_5 :\ \mathbf{r} = \begin{pmatrix}2\\2\\3\end{pmatrix} + s \begin{pmatrix}-1\\-2\\-1\end{pmatrix}$.

Solution. (a) Not parallel. As before, x and y give $\lambda = 3$, $\mu = -2$. But z: $2 + \lambda = 5 \neq 7$. The equations are inconsistent, so l_1 and l_3 are skew.

(b) $\begin{pmatrix}2\\4\\2\end{pmatrix} = 2\begin{pmatrix}1\\2\\1\end{pmatrix}$: parallel. Is O on l_1? $1 + \lambda = 0$ gives $\lambda = -1$, and then $z = 2 + \lambda = 1 \neq 0$. No, so the lines are parallel and distinct.

(c) $\begin{pmatrix}-1\\-2\\-1\end{pmatrix} = -\begin{pmatrix}1\\2\\1\end{pmatrix}$: parallel. Is $(2, 2, 3)$ on l_1? $\lambda = 1$ gives $(2, 2, 3)$. Yes, so l_5 is the same line as l_1.

Common mistake: solving two equations and not checking the third. For l_1 and l_3:

$\lambda = 3,\ \mu = -2$ from x and y, so the lines meet at $(4, 6, 5)$ ×

$z :\ 2 + 3 = 5,$ but $7 + 0(-2) = 7.\ 5 \neq 7,$ so the lines are skew ✓

13.4.6 Angle Involving Lines

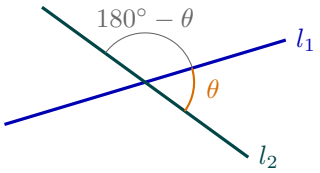
Definition 13.4.8. Acute angle between two lines

Let θ denote the acute angle between $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda\mathbf{b}_1$ and $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu\mathbf{b}_2$. Then

$$\cos\theta = \frac{|\mathbf{b}_1 \cdot \mathbf{b}_2|}{|\mathbf{b}_1|\,|\mathbf{b}_2|}.$$

Only the **directions** enter: the positions $\mathbf{a}_1$ and $\mathbf{a}_2$ do not matter, so the formula works even for skew lines.

? This is slightly different from the original angle formula of Section 13.2.2: there is a modulus sign on top. Why? (Hint: a line has no arrow on it. What happens to $\mathbf{b}_1 \cdot \mathbf{b}_2$ if you replace $\mathbf{b}_2$ by $-\mathbf{b}_2$, which describes the same line?)



Two crossing lines make two angles, θ and $180° - \theta$. The modulus picks the acute one.

Example 13.4.12

Find the acute angle between

(a) $l_1 : \mathbf{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}3\\0\\-1\end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix}3\\-2\\-1\end{pmatrix} + \mu\begin{pmatrix}2\\2\\1\end{pmatrix}$,

(b) $\mathbf{r} = \lambda\begin{pmatrix}1\\2\\2\end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix}1\\0\\0\end{pmatrix} + \mu\begin{pmatrix}-2\\1\\-2\end{pmatrix}$.

Solution. (a) $\cos\theta = \dfrac{|6+0-1|}{\sqrt{10}\sqrt{9}} = \dfrac{5}{3\sqrt{10}}$, so $\theta = 58.2°$ (1 d.p.).

(b) $\cos\theta = \dfrac{|-2+2-4|}{3\times 3} = \dfrac{4}{9}$, so $\theta = 63.6°$ (1 d.p.).

× **Common mistake:** dropping the modulus when the dot product is negative. In (b):

$$\cos\theta = \frac{-4}{9}, \quad \theta = 116.4° \qquad \times$$
$$\cos\theta = \frac{|-4|}{9}, \quad \theta = 63.6° \qquad \checkmark$$

$116.4°$ is the obtuse angle between the lines. The question wants the acute one.

× **Common mistake:** using the position vectors of the lines. In (a):

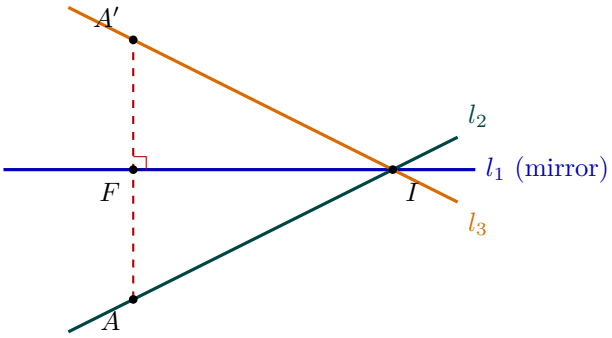
$$\cos\theta = \frac{|(1,2,3)\cdot(3,-2,-1)|}{\sqrt{14}\sqrt{14}} = \frac{4}{14}, \quad \theta = 73.4° \qquad \times$$
$$\cos\theta = \frac{|(3,0,-1)\cdot(2,2,1)|}{\sqrt{10}\sqrt{9}} = \frac{5}{3\sqrt{10}}, \quad \theta = 58.2° \qquad \checkmark$$

13.4.7 Reflection (Line about Line)

Given two intersecting lines, we want the reflection of one line about the other. A line needs a point and a direction, so we need two points of the image line. The intersection point is one of them, because it lies on the mirror line and does not move.

Definition 13.4.9. Method: reflection of l_2 about l_1

1. Find the intersection I of l_1 and l_2 (Section 13.4.5).
2. Take a point A on l_2, other than I. Find the foot F of the perpendicular from A to l_1 (Section 13.4.2).
3. Reflect: $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA}$ (Section 13.4.3).
4. The reflected line passes through I with direction $\overrightarrow{IA'}$.



! The reflection of l_2 about l_1 is **not** the same as the reflection of l_1 about l_2: the mirror is the line you reflect *about*. And if $l_2 \parallel l_1$, there is no intersection: reflect one point of l_2, and keep the same direction.

Example 13.4.13

Find the reflection of l_2 about l_1, where

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \ \lambda \in \mathbb{R}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \ \mu \in \mathbb{R}.$$

Solution. **Step 1: intersection.** $\begin{pmatrix} 1+3\lambda \\ 2 \\ 3-\lambda \end{pmatrix} = \begin{pmatrix} 3+2\mu \\ -2+2\mu \\ -1+\mu \end{pmatrix}$. The y-row gives $\mu = 2$, the z-row then gives $\lambda = 2$, and the x-row checks: $7 = 7$. So I is $\begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix}$.

Step 2: a point on l_2 and its foot. $\mu = 0$ gives $A(3, -2, -1)$ on l_2. Its foot on l_1 is $F(4, 2, 2)$ (Example 13.4.3).

Step 3: reflected point. $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA} = \begin{pmatrix} 5 \\ 6 \\ 5 \end{pmatrix}$ (Example 13.4.5).

Step 4: direction. $\overrightarrow{IA'} = \begin{pmatrix}5\\6\\5\end{pmatrix} - \begin{pmatrix}7\\2\\1\end{pmatrix} = \begin{pmatrix}-2\\4\\4\end{pmatrix} = 2\begin{pmatrix}-1\\2\\2\end{pmatrix}$. The reflected line is

$$l_3 : \mathbf{r} = \begin{pmatrix}7\\2\\1\end{pmatrix} + t\begin{pmatrix}-1\\2\\2\end{pmatrix}, \ t \in \mathbb{R}.$$

Check: l_2 and l_3 make the same angle with l_1, since $\dfrac{|(2,2,1)\cdot(3,0,-1)|}{3} = \dfrac{|(-1,2,2)\cdot(3,0,-1)|}{3} = \dfrac{5}{3}$.

Common mistake: using the foot F instead of the reflected point A' for the direction.

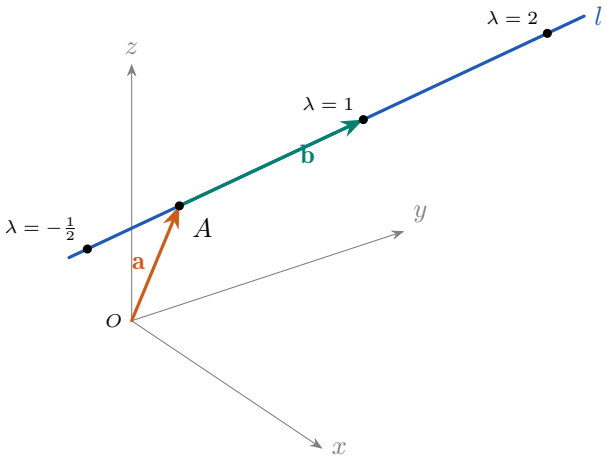
$$\overrightarrow{IF} = \begin{pmatrix}4\\2\\2\end{pmatrix} - \begin{pmatrix}7\\2\\1\end{pmatrix} = \begin{pmatrix}-3\\0\\1\end{pmatrix}, \quad l_3 : \mathbf{r} = \begin{pmatrix}7\\2\\1\end{pmatrix} + t\begin{pmatrix}-3\\0\\1\end{pmatrix} \qquad \times$$

$$\overrightarrow{IA'} = \begin{pmatrix}5\\6\\5\end{pmatrix} - \begin{pmatrix}7\\2\\1\end{pmatrix} = 2\begin{pmatrix}-1\\2\\2\end{pmatrix}, \quad l_3 : \mathbf{r} = \begin{pmatrix}7\\2\\1\end{pmatrix} + t\begin{pmatrix}-1\\2\\2\end{pmatrix} \qquad \checkmark$$

F lies on the mirror, so $\overrightarrow{IF}$ is parallel to l_1: the "answer" is l_1 itself.

13.4.8 Visual Atlas: Lines

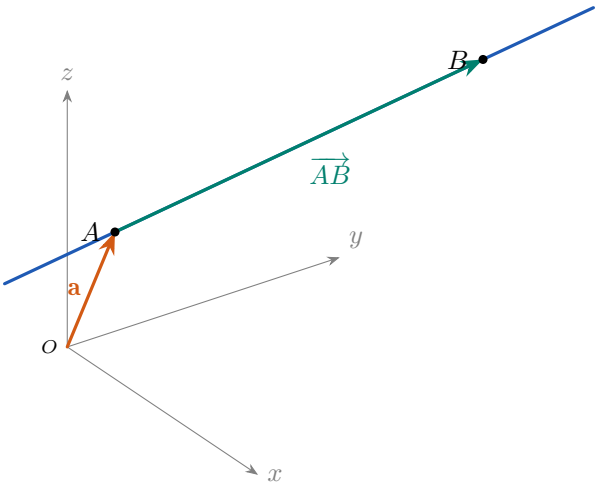
W1.1 $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$



$$l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \ \lambda \in \mathbb{R}$$

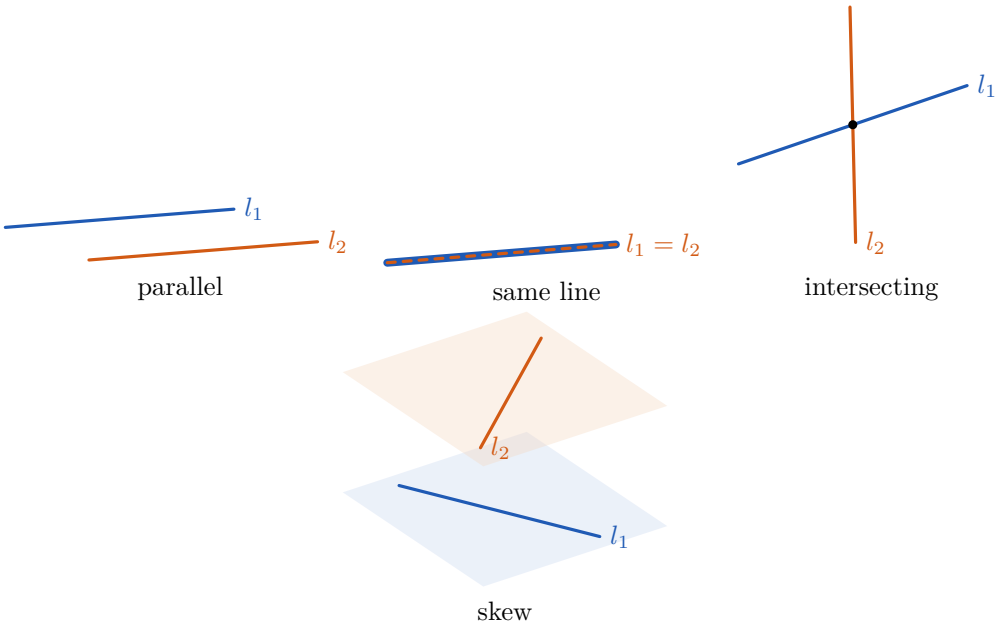
$$\frac{x-1}{1} = \frac{y}{2} = \frac{z-2}{1}$$

W1.2 Line through two points



$$\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}), \ \lambda \in \mathbb{R}$$

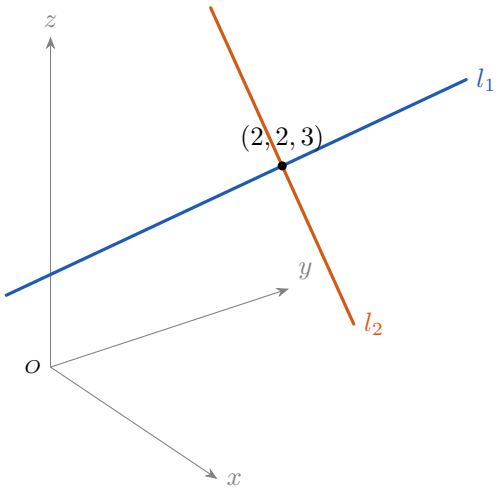
W1.3 Line–line: parallel, same line, intersecting, skew



$$\mathbf{b}_1 = k\mathbf{b}_2,\ A_1 \notin l_2 \Rightarrow \text{parallel}$$
$$\mathbf{b}_1 = k\mathbf{b}_2,\ A_1 \in l_2 \Rightarrow \text{same line}$$
$$\mathbf{b}_1 \neq k\mathbf{b}_2,\ \text{unique } (\lambda, \mu) \Rightarrow \text{intersecting}$$
$$\mathbf{b}_1 \neq k\mathbf{b}_2,\ \text{no } (\lambda, \mu) \Rightarrow \text{skew}$$

Example 13.4.14

$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$. Show l_1, l_2 intersect; find the point and the angle.



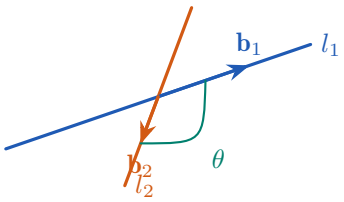
Solution.

$$\begin{pmatrix}1\\2\\1\end{pmatrix} \neq k \begin{pmatrix}1\\0\\-1\end{pmatrix} \quad \therefore \text{not parallel}$$

$$1+\lambda = 1+\mu \quad (1)$$
$$2\lambda = 2 \quad (2)$$
$$2+\lambda = 4-\mu \quad (3)$$
$$\text{From (1), (2): } \lambda = 1,\ \mu = 1$$
$$\text{(3): LHS} = 3,\ \text{RHS} = 4-1 = 3$$
$$\therefore \text{LHS} = \text{RHS}$$
$$\therefore \text{intersect at } (2,2,3)$$

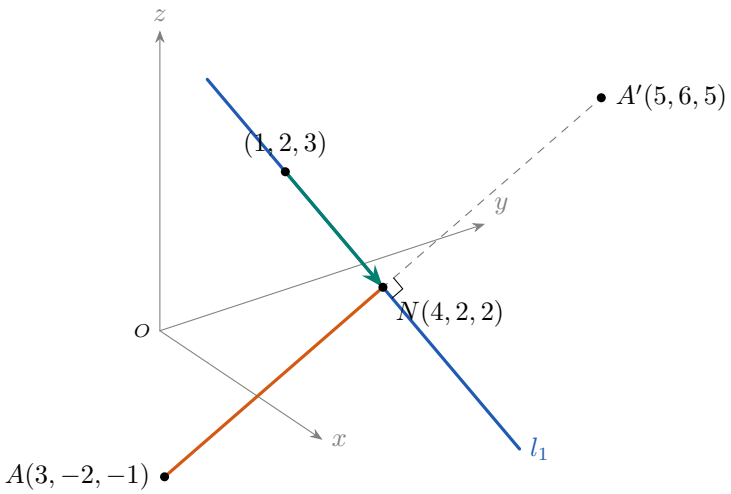
$$\cos\theta = \frac{\left|\begin{pmatrix}1\\2\\1\end{pmatrix} \cdot \begin{pmatrix}1\\0\\-1\end{pmatrix}\right|}{\sqrt{6}\sqrt{2}}$$
$$= 0$$
$$\theta = 90^\circ$$

W1.4 Angle between two lines (acute)



$$\cos\theta = \frac{|\mathbf{b}_1 \cdot \mathbf{b}_2|}{|\mathbf{b}_1||\mathbf{b}_2|}$$

W1.5 Foot of perpendicular, reflection, distance: point and line



$$\overrightarrow{ON} = \begin{pmatrix} 1+3\lambda \\ 2 \\ 3-\lambda \end{pmatrix}$$

$$\overrightarrow{AN} = \begin{pmatrix} -2+3\lambda \\ 4 \\ 4-\lambda \end{pmatrix}$$

$$\overrightarrow{AN} \cdot \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = 0$$

$$10\lambda - 10 = 0$$

$$\lambda = 1$$

$$\therefore N = (4, 2, 2)$$

$$\overrightarrow{ON} = \frac{\overrightarrow{OA} + \overrightarrow{OA'}}{2}$$

$$\overrightarrow{OA'} = 2\overrightarrow{ON} - \overrightarrow{OA}$$

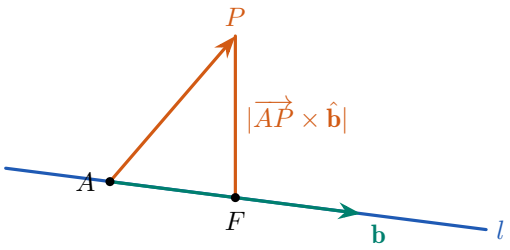
$$= 2\begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ 6 \\ 5 \end{pmatrix}$$

$$AN = \left| \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix} \right|$$

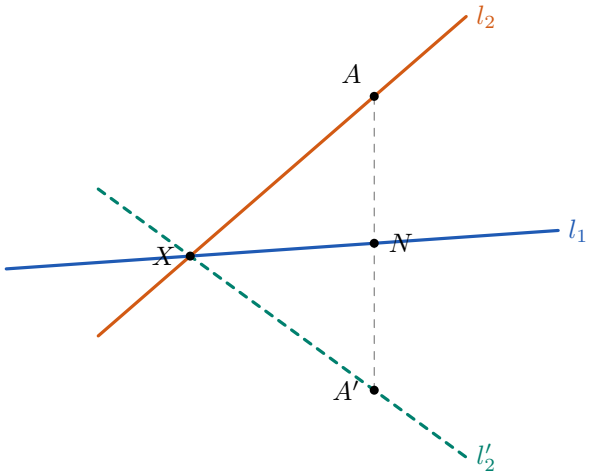
$$= \sqrt{26}$$

W1.6 Distance point to line by cross product



$$PF = |\overrightarrow{AP} \times \hat{\mathbf{b}}|$$

W1.7 Reflection of a line in a line



$X = l_1 \cap l_2$

$A \in l_2 \Rightarrow N$ foot on $l_1 \Rightarrow \overrightarrow{OA'} = 2\overrightarrow{ON} - \overrightarrow{OA}$

$l_2' : \mathbf{r} = \overrightarrow{OX} + \mu\overrightarrow{XA'}$

13.5 Planes

13.5.1 Equations of Planes

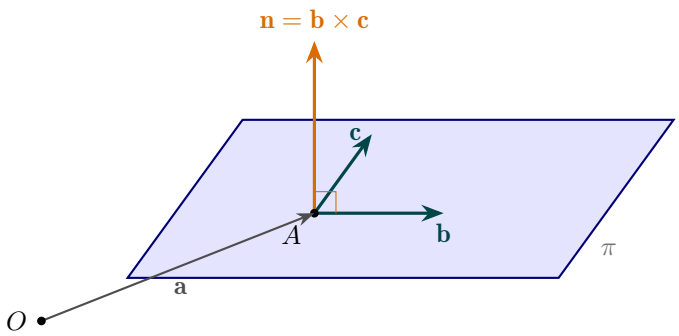
A plane is fixed by **one point on it** and **one normal vector n**, a vector perpendicular to every line in the plane. Equivalently, by one point and two non-parallel directions in the plane.

Definition 13.5.1. Three forms of the equation of a plane

Vector form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}, \quad \lambda, \mu \in \mathbb{R},$
where $\mathbf{a}$ is the position vector of any point on the plane, and $\mathbf{b}$ and $\mathbf{c}$ are any two non-parallel directions parallel to the plane.
Scalar product form: $\mathbf{r} \cdot \mathbf{n} = d,$
where $\mathbf{n}$ is a normal vector of the plane and d is a scalar constant.
Cartesian form: $ax + by + cz = d,$
where a, b, c are scalar constants, and $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is a normal.

From vector form to scalar product form. The normal is perpendicular to both directions, so take $\mathbf{n} = \mathbf{b} \times \mathbf{c}$. Every point $\mathbf{r}$ of the plane satisfies $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$, so

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} \qquad \text{(i.e. } d = \mathbf{a} \cdot \mathbf{n}\text{)}.$$



In Example 13.5.1(a), $A = (1, 2, 3)$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, so $\mathbf{n} = -3\mathbf{i} + 3\mathbf{k}$, or simply $-\mathbf{i} + \mathbf{k}$: any non-zero multiple of a normal is a normal.

Example 13.5.1

Find the Cartesian equation of the plane that passes through

(a) the point $(1, 2, 3)$, and is parallel to $\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$,

(b) the point $A(3, 1, -1)$, and contains the line $\mathbf{r} = \lambda(2\mathbf{i} - \mathbf{j} + \mathbf{k})$, $\lambda \in \mathbb{R}$,

(c) the points $B(0, 1, 1)$, $C(1, 1, 3)$ and $D(2, 0, 2)$.

Solution. (a) A normal is perpendicular to both directions:

$$\mathbf{n} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 - 1 \\ -(2 - 2) \\ 1 + 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 3 \end{pmatrix}, \quad \text{so take } \mathbf{n} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

$$\mathbf{r} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 2, \quad \text{i.e. } -x + z = 2.$$

(b) The line passes through O, so the plane contains O and A, and $\overrightarrow{OA}$ is a second direction:

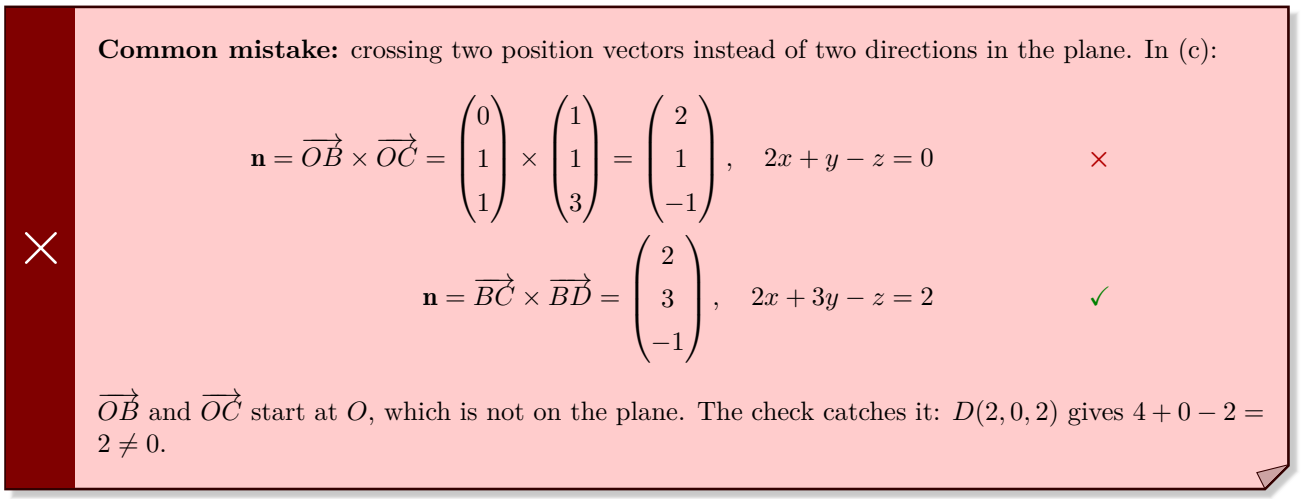
$$\mathbf{n} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1-1 \\ -(3+2) \\ -3-2 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \\ -5 \end{pmatrix}, \quad \text{so take } \mathbf{n} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

O is on the plane, so $d = \mathbf{0} \cdot \mathbf{n} = 0$, i.e. $y + z = 0$.

(c) Two directions in the plane are $\overrightarrow{BC} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\overrightarrow{BD} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

$$\mathbf{n} = \overrightarrow{BC} \times \overrightarrow{BD} = \begin{pmatrix} 0+2 \\ -(1-4) \\ -1-0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}, \qquad d = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = 2,$$

i.e. $2x + 3y - z = 2$. Check D: $4 + 0 - 2 = 2$.

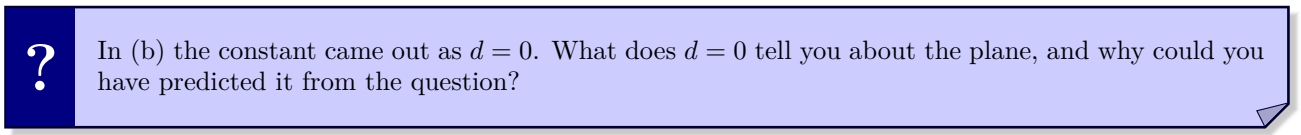


Common mistake: crossing two position vectors instead of two directions in the plane. In (c):

$$\mathbf{n} = \overrightarrow{OB} \times \overrightarrow{OC} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \quad 2x + y - z = 0 \qquad \times$$

$$\mathbf{n} = \overrightarrow{BC} \times \overrightarrow{BD} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}, \quad 2x + 3y - z = 2 \qquad \checkmark$$

$\overrightarrow{OB}$ and $\overrightarrow{OC}$ start at O, which is not on the plane. The check catches it: $D(2, 0, 2)$ gives $4 + 0 - 2 = 2 \neq 0$.



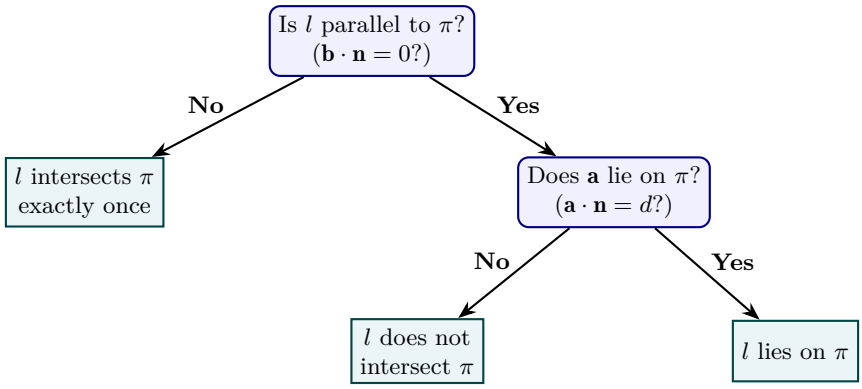
In (b) the constant came out as $d = 0$. What does $d = 0$ tell you about the plane, and why could you have predicted it from the question?

13.5.2 Relationship Between Line and Plane

Given a line l and a plane π, exactly one of these is true:

1. l lies on π;
2. l does not intersect π (it is parallel to π, off the plane);
3. l intersects π exactly once (most common).

To identify which case it is, always draw the diagram, then follow these steps for $l:\ \mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$ and $\pi:\ \mathbf{r}\cdot\mathbf{n}=d$.



Example 13.5.2

The plane π has equation $\mathbf{r}\cdot\begin{pmatrix}2\\-1\\2\end{pmatrix}=6$. Determine the relationship between π and each line.

$$l_a:\ \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\0\end{pmatrix},\qquad l_b:\ \mathbf{r}=\lambda\begin{pmatrix}1\\2\\0\end{pmatrix},\qquad l_c:\ \mathbf{r}=\begin{pmatrix}1\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\1\end{pmatrix}.$$

Solution. l_a: $\mathbf{b}\cdot\mathbf{n}=2-2+0=0$, so $l_a\parallel\pi$. $\mathbf{a}\cdot\mathbf{n}=2+0+4=6=d$, so l_a lies on π.

l_b: same direction, so $l_b\parallel\pi$. $\mathbf{a}\cdot\mathbf{n}=0\neq 6$, so l_b does not intersect π.

l_c: $\mathbf{b}\cdot\mathbf{n}=2-1+2=3\neq 0$, so l_c intersects π exactly once. Substituting, $2(1+\lambda)-\lambda+2\lambda=6$, so $\lambda=\dfrac{4}{3}$ and the point is $\begin{pmatrix}\frac{7}{3}\\\frac{4}{3}\\\frac{4}{3}\end{pmatrix}$.

Intersection Between Line and Plane

The general point of the line, $\mathbf{a}+\lambda\mathbf{b}$, is on the plane exactly when it satisfies the plane's equation. Substitute it into $\mathbf{r}\cdot\mathbf{n}=d$, solve for λ, and substitute back.

Example 13.5.3

Find the coordinates of the point of intersection between the plane $\mathbf{r}\cdot(\mathbf{i}+2\mathbf{j}+\mathbf{k})=5$ and the line $\mathbf{r}=(\mathbf{i}+3\mathbf{j}+\mathbf{k})+\lambda(\mathbf{i}-2\mathbf{j}+\mathbf{k})$, $\lambda\in\mathbb{R}$.

Solution. $\begin{pmatrix} 1+\lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 5$: $1+\lambda+6-4\lambda+1+\lambda = 8-2\lambda = 5$, so $\lambda = \frac{3}{2}$.

The position vector is $\begin{pmatrix} \frac{5}{2} \\ 0 \\ \frac{5}{2} \end{pmatrix}$, so the point is $\begin{pmatrix} \frac{5}{2} \\ 0 \\ \frac{5}{2} \end{pmatrix}$, i.e. the coordinates are $\left(\frac{5}{2}, 0, \frac{5}{2}\right)$.

Common mistake: reading $\mathbf{b} \cdot \mathbf{n} = 0$ as "the line is perpendicular to the plane". For l_a in Example 13.5.2:

$$\mathbf{b} \cdot \mathbf{n} = 0, \text{ so } l_a \perp \pi \quad \times$$
$$\mathbf{b} \cdot \mathbf{n} = 0, \text{ so } \mathbf{b} \perp \mathbf{n}, \text{ and } l_a \parallel \pi \quad \checkmark$$

The line is perpendicular to the *normal*, so it runs along the plane. A line perpendicular to π has $\mathbf{b} \parallel \mathbf{n}$.

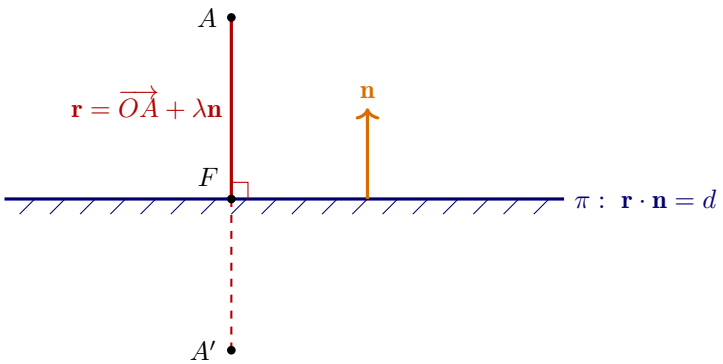
Take note of whether the question asks for the answer as **coordinates** $\left(\frac{5}{2}, 0, \frac{5}{2}\right)$ or as a **position vector** $\frac{5}{2}\mathbf{i} + \frac{5}{2}\mathbf{k}$, and answer accordingly.

13.5.3 Foot of Perpendicular (Point to Plane)

The perpendicular from a point A to a plane runs along the **normal**. So the foot is where the line through A in the direction $\mathbf{n}$ meets the plane: a line–plane intersection (Section 13.5.2).

Definition 13.5.2. Method: foot of perpendicular from A to π : $\mathbf{r}\cdot\mathbf{n}=d$

1. Write the line through A perpendicular to π: l_{AF} : $\mathbf{r}=\overrightarrow{OA}+\lambda\mathbf{n}$.
2. F lies on π: substitute $\overrightarrow{OF}=\overrightarrow{OA}+\lambda\mathbf{n}$ into $\mathbf{r}\cdot\mathbf{n}=d$, and solve for λ.
3. Substitute λ back into $\overrightarrow{OF}$.



Example 13.5.4

Find the foot of perpendicular of the point $(7,0,3)$ to the plane p with equation $\mathbf{r}\cdot(\mathbf{i}+\mathbf{j}+2\mathbf{k})=1$.

Solution. Line through $A(7,0,3)$ along the normal: $\mathbf{r}=\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}$. On p:

$$\begin{pmatrix}7+\lambda\\ \lambda\\ 3+2\lambda\end{pmatrix}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=7+\lambda+\lambda+6+4\lambda=13+6\lambda=1,\quad \lambda=-2.$$

$\overrightarrow{OF}=\begin{pmatrix}7\\0\\3\end{pmatrix}-2\begin{pmatrix}1\\1\\2\end{pmatrix}$, so the foot is $\begin{pmatrix}5\\-2\\-1\end{pmatrix}$, i.e. $(5,-2,-1)$.

Example 13.5.5

Find the coordinates of the foot of perpendicular from the point $Q(1,1,0)$ to the plane p with equation $\mathbf{r}\cdot(\mathbf{i}+2\mathbf{j}-\mathbf{k})=2$.

Solution. $\mathbf{r} = \begin{pmatrix}1\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\2\\-1\end{pmatrix}$. On p: $(1+\lambda)+2(1+2\lambda)-(-\lambda) = 3+6\lambda = 2$, so $\lambda = -\frac{1}{6}$.

$$\overrightarrow{OF} = \begin{pmatrix}1\\1\\0\end{pmatrix} - \frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}\frac{5}{6}\\\frac{2}{3}\\\frac{1}{6}\end{pmatrix}, \quad \text{so the foot is } \begin{pmatrix}\frac{5}{6}\\\frac{2}{3}\\\frac{1}{6}\end{pmatrix}, \text{ i.e. } \left(\frac{5}{6}, \frac{2}{3}, \frac{1}{6}\right).$$

Check: $\frac{5}{6} + \frac{4}{3} - \frac{1}{6} = 2$. ✓

Common mistake: a sign slip when multiplying out $-\frac{1}{6}\mathbf{n}$.

$$\overrightarrow{OF} = \begin{pmatrix}1\\1\\0\end{pmatrix} - \frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}\frac{5}{6}\\\frac{4}{3}\\-\frac{1}{6}\end{pmatrix} \quad \times$$

$$\overrightarrow{OF} = \begin{pmatrix}1\\1\\0\end{pmatrix} - \frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}\frac{5}{6}\\\frac{2}{3}\\\frac{1}{6}\end{pmatrix} \quad ✓$$

Always substitute the foot back into the plane: $\frac{5}{6} + \frac{8}{3} + \frac{1}{6} = \frac{11}{3} \neq 2$ exposes the first answer at once.

Reflection (Point about Plane)

This is the same as Section 13.4.3: the foot is the midpoint of A and its image, so $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA}$.

Example 13.5.6

Find the reflections of $A(7,0,3)$ and of $Q(1,1,0)$ in the planes of Examples 13.5.4 and 13.5.5.

Solution. $\overrightarrow{OA'} = 2\begin{pmatrix}5\\-2\\-1\end{pmatrix} - \begin{pmatrix}7\\0\\3\end{pmatrix} = \begin{pmatrix}3\\-4\\-5\end{pmatrix}$, $\overrightarrow{OQ'} = 2\begin{pmatrix}\frac{5}{6}\\\frac{2}{3}\\\frac{1}{6}\end{pmatrix} - \begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}\frac{2}{3}\\\frac{1}{3}\\\frac{1}{3}\end{pmatrix}$.

13.5.4 Shortest Distance (Point to Plane)

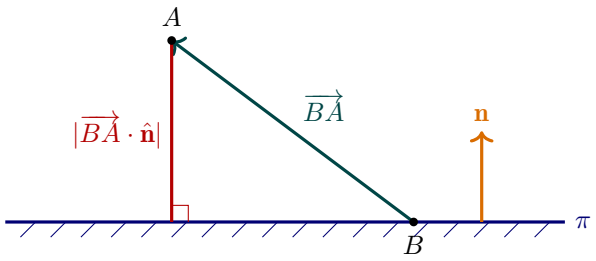
Definition 13.5.3. Shortest distance from A to π : $\mathbf{r}\cdot\mathbf{n}=d$

Let B be any point on π. Then

$$\text{Shortest distance} = \left|\overrightarrow{BA}\cdot\hat{\mathbf{n}}\right| = \frac{\left|\overrightarrow{BA}\cdot\mathbf{n}\right|}{|\mathbf{n}|}.$$

Since $\overrightarrow{OB}\cdot\mathbf{n}=d$, no explicit B is needed:

$$\text{Shortest distance} = \frac{\left|(\overrightarrow{OA}-\overrightarrow{OB})\cdot\mathbf{n}\right|}{|\mathbf{n}|} = \frac{\left|\overrightarrow{OA}\cdot\mathbf{n}-d\right|}{|\mathbf{n}|}.$$



Why $|\mathbf{a}\cdot\hat{\mathbf{n}}|$, and not $|\mathbf{a}\times\hat{\mathbf{n}}|$? For a line we used the cross product, because the distance is the side *perpendicular* to the line's direction. Here the vector in consideration is the *normal*, and the distance is the side *along* the normal: the adjacent side, $|\overrightarrow{BA}|\cos\theta$. Adjacent means dot.
We can use the foot of perpendicular too (then the distance is AF), but as with lines, it is unnecessary and tedious.

Example 13.5.7

Find the shortest distance of the point $(7,0,3)$ to the plane p with equation $\mathbf{r}\cdot(\mathbf{i}+\mathbf{j}+2\mathbf{k})=1$.

Solution. $B(1,0,0)$ lies on p. $\overrightarrow{BA} = \begin{pmatrix}7\\0\\3\end{pmatrix} - \begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}6\\0\\3\end{pmatrix}$.

$$\text{Distance} = \frac{\left|\begin{pmatrix}6\\0\\3\end{pmatrix}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\1\\2\end{pmatrix}\right|} = \frac{12}{\sqrt{6}} = 2\sqrt{6}\text{ units.}$$

Check with Example 13.5.4: $|\overrightarrow{AF}| = \left|\begin{pmatrix}-2\\-2\\-4\end{pmatrix}\right| = \sqrt{24} = 2\sqrt{6}$.

Example 13.5.8

Find the shortest distance from the point $Q(1,1,0)$ to the plane p with equation $\mathbf{r}\cdot(\mathbf{i}+2\mathbf{j}-\mathbf{k})=2$.

Solution. **With a point.** $B(0,1,0)$ lies on p. $\overrightarrow{BQ} = \begin{pmatrix}1\\0\\0\end{pmatrix}$, and the distance is $\dfrac{\left|\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-1\end{pmatrix}\right|}{\sqrt{6}} = \dfrac{1}{\sqrt{6}}$ units.

Without a point. Let B be *any* point on p, so $\overrightarrow{OB}\cdot\mathbf{n} = 2$. Then

$$\text{Distance} = \frac{\left|\overrightarrow{OQ}\cdot\mathbf{n} - \overrightarrow{OB}\cdot\mathbf{n}\right|}{|\mathbf{n}|} = \frac{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-1\end{pmatrix} - 2\right|}{\sqrt{6}} = \frac{|3-2|}{\sqrt{6}} = \frac{1}{\sqrt{6}} \text{ units.}$$

The second method is useful for abstract vector questions, such as the shortest distance from A to $\mathbf{r}\cdot\mathbf{n} = d$.

Common mistake: using the cross product, as for a line. In Example 13.5.7:

$$\text{Distance} = \frac{\left|\overrightarrow{BA}\times\mathbf{n}\right|}{|\mathbf{n}|} = \frac{\left|\begin{pmatrix}-3\\-9\\6\end{pmatrix}\right|}{\sqrt{6}} = \frac{\sqrt{126}}{\sqrt{6}} = \sqrt{21} \qquad \times$$

$$\text{Distance} = \frac{\left|\overrightarrow{BA}\cdot\mathbf{n}\right|}{|\mathbf{n}|} = \frac{12}{\sqrt{6}} = 2\sqrt{6} \qquad \checkmark$$

$\sqrt{21}$ is the distance from B to F, *along* the plane.

How do we find the shortest distance between a line and a plane, when the line does not lie on the plane? Which case of Section 13.5.2 must the line be in for the question to make sense, and why may any point of the line then be used?

Example 13.5.9

Show that the line $l:\ \mathbf{r} = \begin{pmatrix}7\\0\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\-1\end{pmatrix}$ is parallel to $p:\ \mathbf{r}\cdot(\mathbf{i}+\mathbf{j}+2\mathbf{k}) = 1$, and find the distance between them.

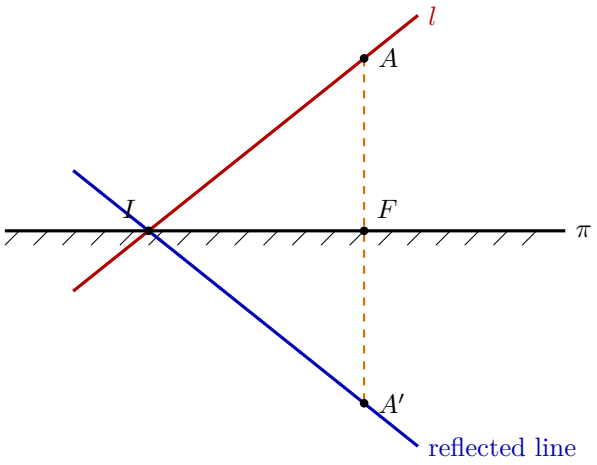
Solution. $\begin{pmatrix}1\\1\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix} = 1+1-2 = 0$, and $(7,0,3)$ does not satisfy the equation $(7+0+6 = 13 \neq 1)$, so $l \parallel p$ and l is off the plane. Every point of l is the same distance from p; use $(7,0,3)$. By Example 13.5.7, the distance is $2\sqrt{6}$ units.

13.5.5 Reflection (Line about Plane)

Given a line and a plane, we want the reflection of the line about the plane. As in Section 13.4.7, we need two points of the image: the intersection with the plane, which does not move, and the image of one other point.

Definition 13.5.4. Steps: reflection of l about π

1. Find the intersection I between l and π (Section 13.5.2).
2. Find the foot of perpendicular F of a point A on l onto π (Section 13.5.3).
3. Find the reflected point: $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA}$ (Section 13.4.3).
4. Find the direction of the line, $\overrightarrow{IA'}$. The reflected line is $\mathbf{r} = \overrightarrow{OI} + \mu\,\overrightarrow{IA'}$.



Example 13.5.10

Find the reflection of the line $l:\ \mathbf{r} = (7\mathbf{i} + 3\mathbf{k}) + \lambda(2\mathbf{i} + \mathbf{k})$, $\lambda \in \mathbb{R}$, about the plane $\pi:\ \mathbf{r} \cdot (\mathbf{i} + \mathbf{j} + 2\mathbf{k}) = 1$.

Solution. ① **Intersection.** $\begin{pmatrix} 7+2\lambda \\ 0 \\ 3+\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 13 + 4\lambda = 1$, so $\lambda = -3$ and $\overrightarrow{OI} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

② **Foot.** $A(7, 0, 3)$ is on l ($\lambda = 0$). Its foot on π is $F(5, -2, -1)$ (Example 13.5.4).

③ **Reflected point.** $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix}$ (Example 13.5.6).

④ **Direction.** $\overrightarrow{IA'} = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix}$. The reflected line is

$$\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix},\ \mu \in \mathbb{R}, \quad \text{i.e. } \mathbf{r} = \mathbf{i} + \mu(2\mathbf{i} - 4\mathbf{j} - 5\mathbf{k}).$$

Example 13.5.11

Find the reflection of the line $l:\ \mathbf{r} = (\mathbf{i} + \mathbf{j}) + \lambda\mathbf{i}$, $\lambda \in \mathbb{R}$, about the plane $\pi:\ \mathbf{r} \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 2$.

Solution. ① $\begin{pmatrix}1+\lambda\\1\\0\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-1\end{pmatrix}=3+\lambda=2$, so $\lambda=-1$ and $\overrightarrow{OI}=\begin{pmatrix}0\\1\\0\end{pmatrix}$.

② $A(1,1,0)$ is on l. Its foot on π is $F\left(\frac{5}{6},\frac{2}{3},\frac{1}{6}\right)$ (Example 13.5.5).

③ $\overrightarrow{OA'}=\begin{pmatrix}\frac{2}{3}\\\frac{1}{3}\\\frac{1}{3}\end{pmatrix}$ (Example 13.5.6).

④ $\overrightarrow{IA'}=\begin{pmatrix}\frac{2}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix}=\frac{1}{3}\begin{pmatrix}2\\-2\\1\end{pmatrix}$. The reflected line is

$$\mathbf{r}=\begin{pmatrix}0\\1\\0\end{pmatrix}+\mu\begin{pmatrix}2\\-2\\1\end{pmatrix},\ \mu\in\mathbb{R},\quad \text{i.e. } \mathbf{r}=\mathbf{j}+\mu(2\mathbf{i}-2\mathbf{j}+\mathbf{k}).$$

> **!** If l is parallel to π, step ① fails: there is no intersection. Then the reflected line is parallel to l. Reflect one point of l, and keep the direction $\mathbf{b}$.

13.5.6 Relationship Between Two Planes

Definition 13.5.5. Two planes $p_1: \mathbf{r}\cdot\mathbf{n}_1 = d_1$ and $p_2: \mathbf{r}\cdot\mathbf{n}_2 = d_2$

1. p_1 and p_2 are not parallel, and therefore intersect in a line (most common): $\mathbf{n}_1 \nparallel \mathbf{n}_2$.
2. p_1 and p_2 are parallel but are not the same plane: after scaling so that $\mathbf{n}_1 = \mathbf{n}_2$, $d_1 \neq d_2$.
3. p_1 and p_2 are the same plane: after scaling so that $\mathbf{n}_1 = \mathbf{n}_2$, $d_1 = d_2$.

Common mistake: comparing the constants before the normals are made equal. For $x+y+z=5$ and $2x+2y+2z=10$:

$\mathbf{n}_1 \parallel \mathbf{n}_2$ and $5 \neq 10$, so the planes are parallel and distinct ×

$2x+2y+2z=10 \iff x+y+z=5$: $\mathbf{n}_1 = \mathbf{n}_2$, $d_1 = d_2$, the same plane ✓

Intersection Between Two Planes

Two non-parallel planes intersect in a line. Two methods find it.

Method 1: by GC (recommended). apps, PlySmlt2, SIMULT EQN SOLVER, EQUATIONS: 2, UNKNOWNS: 3. Enter the two equations and press SOLVE. The GC gives x_1 and x_2 in terms of x_3, with $x_3 = x_3$ free. Let $z = \lambda$ and read off the line.

Method 2: manual. Only necessary if there are unknown constants, e.g. $\mathbf{r}\cdot\begin{pmatrix}1\\2\\a\end{pmatrix} = 1$; then the answer will possibly be in terms of a. Steps: ① the direction is $\mathbf{n}_1 \times \mathbf{n}_2$ (it lies in both planes, so it is perpendicular to both normals); ② let $z=0$ and solve for x and y simultaneously, to get one point.

Example 13.5.12

Find the equation of the line of intersection between the planes $3x+2y+z=10$ and $x-2y+z=5$.

Solution. **Method 1 (GC).** The GC gives $x = \frac{15}{4} - \frac{1}{2}z$, $y = -\frac{5}{8} + \frac{1}{4}z$, $z = z$. Let $z = \lambda$:

$$\mathbf{r} = \begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix} + \lambda\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix},\ \lambda \in \mathbb{R}.$$

Method 2 (manual). ① $\begin{pmatrix}3\\2\\1\end{pmatrix} \times \begin{pmatrix}1\\-2\\1\end{pmatrix} = \begin{pmatrix}2+2\\-(3-1)\\-6-2\end{pmatrix} = \begin{pmatrix}4\\-2\\-8\end{pmatrix} = 2\begin{pmatrix}2\\-1\\-4\end{pmatrix}$. ② Let $z=0$: $3x+2y=10$ and $x-2y=5$ give $x=\frac{15}{4}$, $y=-\frac{5}{8}$. So $\mathbf{r} = \begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix} + \lambda\begin{pmatrix}2\\-1\\-4\end{pmatrix}$.

? Another student's answer is $\mathbf{r} = \begin{pmatrix} 0 \\ \frac{5}{4} \\ \frac{15}{2} \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ -4 \end{pmatrix}$. Is it the same line? What is the quickest way to decide?

Example 13.5.13

Find the equation of the line of intersection between the planes $x + 2y - 2z = 3$ and $3x - 2y + 4z = 5$.

Solution. **Method 1 (GC).** $x = 2 - \frac{1}{2}z,\ y = \frac{1}{2} + \frac{5}{4}z,\ z = z$. So $\mathbf{r} = \begin{pmatrix} 2 \\ \frac{1}{2} \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -\frac{1}{2} \\ \frac{5}{4} \\ 1 \end{pmatrix},\ \mu \in \mathbb{R}$.

Method 2 (manual). $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 8-4 \\ -(4+6) \\ -2-6 \end{pmatrix} = 2\begin{pmatrix} 2 \\ -5 \\ -4 \end{pmatrix}$. With $z = 0$: $x + 2y = 3$, $3x - 2y = 5$, so $x = 2$, $y = \frac{1}{2}$, and $\mathbf{r} = \begin{pmatrix} 2 \\ \frac{1}{2} \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -5 \\ -4 \end{pmatrix}$.

13.5.7 Angles Involving Planes

Angle Between Two Planes

Definition 13.5.6. Acute angle between two planes

Let θ denote the acute angle between $p_1 : \mathbf{r} \cdot \mathbf{n}_1 = d_1$ and $p_2 : \mathbf{r} \cdot \mathbf{n}_2 = d_2$. Then

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1|\,|\mathbf{n}_2|}.$$

The angle between two planes equals the angle between their normals. The modulus picks the acute angle, exactly as for two lines.

Example 13.5.14

Find the angle between the planes $3x - y + 2z = 3$ and $y = 2$.

Solution. $y = 2$ has normal $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. $\cos\theta = \dfrac{\left|\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right|}{\sqrt{14}\,(1)} = \dfrac{1}{\sqrt{14}}$, so $\theta = 74.5°$ (1 d.p.).

Example 13.5.15

Find the angle between the planes $2x - 3y + 4z = 6$ and $x + 2y - 5z = 8$.

Solution. $\cos\theta = \dfrac{\left|\begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix}\right|}{\sqrt{29}\,\sqrt{30}} = \dfrac{|2 - 6 - 20|}{\sqrt{870}} = \dfrac{24}{\sqrt{870}}$, so $\theta = 35.5°$ (1 d.p.).

Common mistake: leaving out the modulus, which gives the obtuse angle.

$$\cos\theta = \frac{-24}{\sqrt{870}}, \quad \theta = 144.5° \qquad \times$$

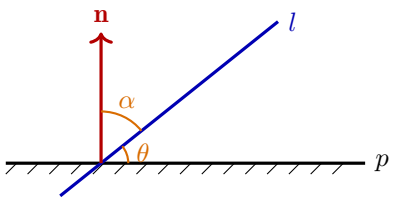
$$\cos\theta = \frac{|-24|}{\sqrt{870}}, \quad \theta = 35.5° \qquad \checkmark$$

Angle Between Line and Plane

Definition 13.5.7. Acute angle between a line and a plane

Let θ denote the acute angle between $l : \mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and $p : \mathbf{r} \cdot \mathbf{n} = d$. Then

$$\sin\theta = \frac{|\mathbf{n} \cdot \mathbf{b}|}{|\mathbf{n}|\,|\mathbf{b}|}.$$



The dot product $\mathbf{n} \cdot \mathbf{b}$ measures the angle α between the line and the *normal*. The angle with the plane is $\theta = 90° - \alpha$, and $\cos\alpha = \cos(90° - \theta) = \sin\theta$.

! **Note that although this formula has "$\sin\theta$" rather than "$\cos\theta$", it is still the DOT product.** The "sin" is there rather than "cos" to account for the $90° - \alpha$ step at the back. You may use Method 2 (find α with cos, then $\theta = 90° - \alpha$) if you are uncomfortable with this.

Example 13.5.16

Find the angle between $\mathbf{r} = \begin{pmatrix} -2 \\ -4 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix}$, $\mu \in \mathbb{R}$, and $\mathbf{r} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 1$.

Solution. **Method 1 ($\sin\theta$ directly).** $\sin\theta = \dfrac{\left| \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right|}{\sqrt{50}\,(1)} = \dfrac{5}{\sqrt{50}} = \dfrac{1}{\sqrt{2}}$, so $\theta = 45°$.

Method 2 ($\cos\alpha$, then $90° - \alpha$). $\cos\alpha = \dfrac{1}{\sqrt{2}}$, so $\alpha = 45°$ and $\theta = 90° - 45° = 45°$.

Example 13.5.17

Find the angle between $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}$, $\mu \in \mathbb{R}$, and $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} = 4$.

Solution. **Method 1.** $\sin\theta = \dfrac{\left| \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \right|}{7 \times 3} = \dfrac{|4 - 6 - 6|}{21} = \dfrac{8}{21}$, so $\theta = 22.4°$ (1 d.p.).

Method 2. $\cos\alpha = \dfrac{8}{21}$, so $\alpha = 67.6°$ and $\theta = 90° - \alpha = 22.4°$ (1 d.p.).

× **Common mistake:** using cos and stopping, so the answer is the angle with the normal.

$$\cos\theta = \frac{8}{21}, \quad \theta = 67.6° \qquad \times$$

$$\sin\theta = \frac{8}{21}, \quad \theta = 22.4° \quad (\text{or } 90° - 67.6°) \qquad \checkmark$$

Draw the diagram to see which angle the question wants.

13.5.8 Distance Between Parallel Planes

Definition 13.5.8. Distance from a plane to the origin, and between parallel planes

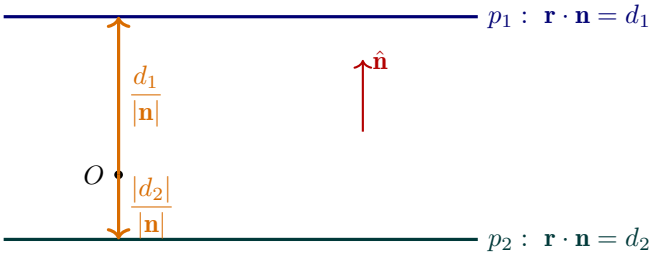
The distance from the plane $\pi:\ \mathbf{r}\cdot\mathbf{n}=k$ to the origin is

$$\text{Distance between } \pi \text{ and the origin} = \left|\frac{k}{|\mathbf{n}|}\right|.$$

So for two parallel planes $p_1:\ \mathbf{r}\cdot\mathbf{n}=d_1$ and $p_2:\ \mathbf{r}\cdot\mathbf{n}=d_2$ with **the same n**,

$$\text{Distance between } p_1 \text{ and } p_2 = \left|\frac{d_1}{|\mathbf{n}|}-\frac{d_2}{|\mathbf{n}|}\right|.$$

The first formula is Section 13.5.4 with $A=O$: $\dfrac{|\mathbf{0}\cdot\mathbf{n}-k|}{|\mathbf{n}|}$. The second subtracts the two *signed* distances from O, measured along the same unit normal $\hat{\mathbf{n}}$: when the planes are on opposite sides of O, the distances add.



? What does it mean if a plane p has equation $\mathbf{r}\cdot\mathbf{n}=0$?

Example 13.5.18

Find the distance between the planes $x+y+z=5$ and $x+y+z=-3$.

Solution. Same normal $\begin{pmatrix}1\\1\\1\end{pmatrix}$, of length $\sqrt{3}$. Distance $=\left|\dfrac{5}{\sqrt{3}}-\dfrac{-3}{\sqrt{3}}\right|=\dfrac{8}{\sqrt{3}}$ units.

Example 13.5.19

Find the distance between the planes $x+2y-3z=2$ and $2x+4y-6z=-10$.

Solution. Make the normals the same first: $2x+4y-6z=-10 \iff \mathbf{r}\cdot\begin{pmatrix}1\\2\\-3\end{pmatrix}=-5$. With $\mathbf{r}\cdot\begin{pmatrix}1\\2\\-3\end{pmatrix}=2$,

$$\text{Distance} = \left|\frac{2}{\sqrt{14}}-\frac{-5}{\sqrt{14}}\right| = \frac{7}{\sqrt{14}} \text{ units.}$$

Common mistake: subtracting the constants before the normals are the same.

$$\text{Distance} = \frac{|2-(-10)|}{\sqrt{14}} = \frac{12}{\sqrt{14}} \quad \times$$

$$\text{Distance} = \frac{|2-(-5)|}{\sqrt{14}} = \frac{7}{\sqrt{14}} \quad \checkmark$$

Example 13.5.20

The plane π has equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 4$. Find the possible Cartesian equations of the plane π' such that π' is parallel to π and the perpendicular distance between π and π' is $\frac{5\sqrt{2}}{2}$ units.

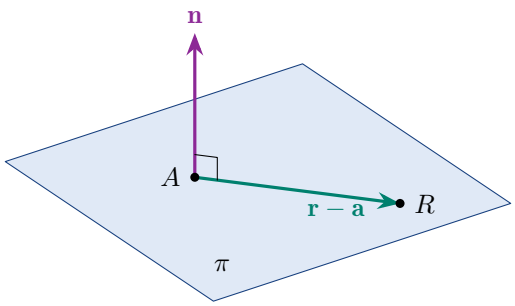
Solution. Draw π' on either side of π. Let $\pi' : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = k$. Then

$$\left| \frac{k}{\sqrt{2}} - \frac{4}{\sqrt{2}} \right| = \frac{5\sqrt{2}}{2} \Rightarrow \frac{|k-4|}{\sqrt{2}} = \frac{5}{\sqrt{2}} \Rightarrow |k-4| = 5 \Rightarrow k = 9 \text{ or } k = -1.$$

So π' is $x + z = 9$ or $x + z = -1$.

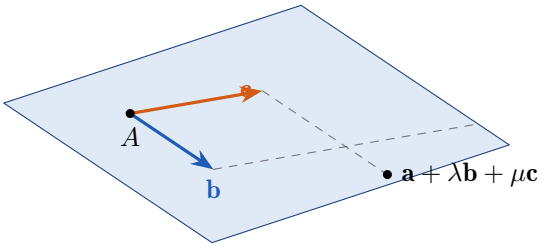
13.5.9 Visual Atlas: Planes

W2.1 Scalar product form: point and normal



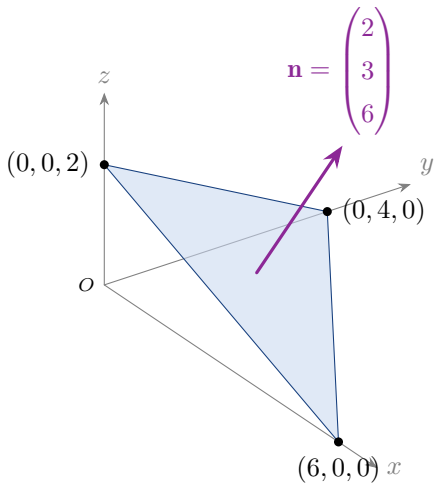
$$(\mathbf{r}-\mathbf{a})\cdot\mathbf{n}=0$$
$$\mathbf{r}\cdot\mathbf{n}=\mathbf{a}\cdot\mathbf{n}$$
$$\mathbf{r}\cdot\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=d \iff n_1x+n_2y+n_3z=d$$

W2.2 Parametric form: point and two directions



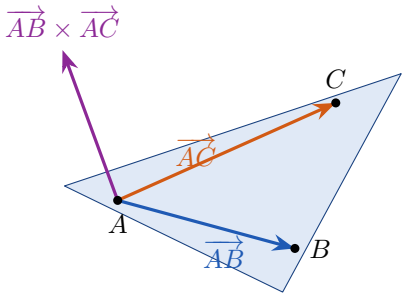
$$\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c},\ \ \lambda,\mu\in\mathbb{R}$$
$$\mathbf{n}=\mathbf{b}\times\mathbf{c}$$

W2.3 Cartesian form and intercepts: $2x+3y+6z=12$



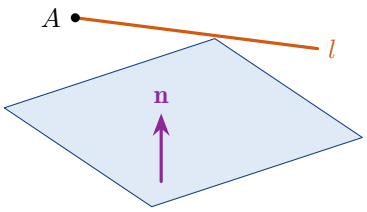
$$y = z = 0 \Rightarrow x = 6$$
$$x = z = 0 \Rightarrow y = 4$$
$$x = y = 0 \Rightarrow z = 2$$

W2.4 Plane through three points

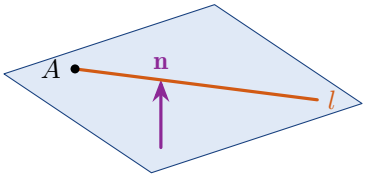


$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}$$
$$\mathbf{r} \cdot \mathbf{n} = \overrightarrow{OA} \cdot \mathbf{n}$$

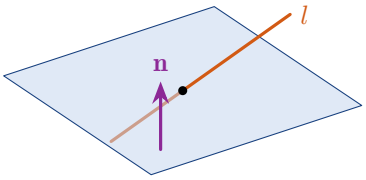
W2.5 Line–plane: three cases



$\mathbf{b} \cdot \mathbf{n} = 0$, $\mathbf{a} \cdot \mathbf{n} \neq d$



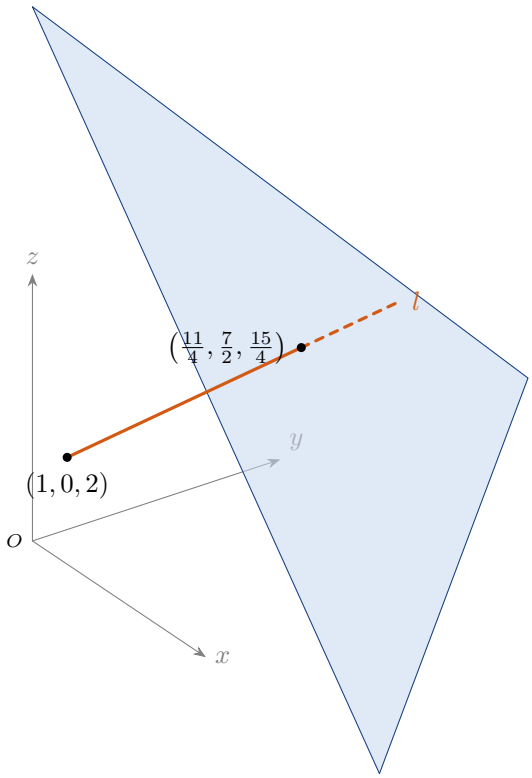
$\mathbf{b} \cdot \mathbf{n} = 0$, $\mathbf{a} \cdot \mathbf{n} = d$



$\mathbf{b} \cdot \mathbf{n} \neq 0$: one point

Example 13.5.21

$l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $p : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 10$. Find the point of intersection and the acute angle between l and p.



Solution.

$$\begin{pmatrix} 1+\lambda \\ 2\lambda \\ 2+\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 10$$

$$3 + 4\lambda = 10$$

$$\lambda = \frac{7}{4}$$

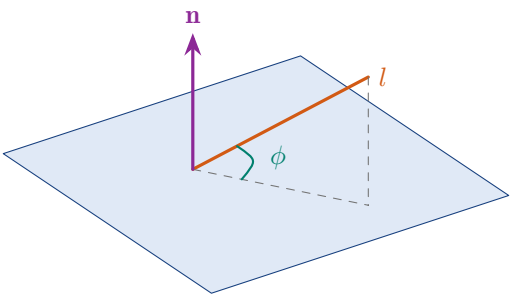
$$\therefore \text{point } \left(\frac{11}{4}, \frac{7}{2}, \frac{15}{4}\right)$$

$$\sin\phi = \frac{\left|\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right|}{\sqrt{6}\sqrt{3}}$$

$$= \frac{4}{\sqrt{18}}$$

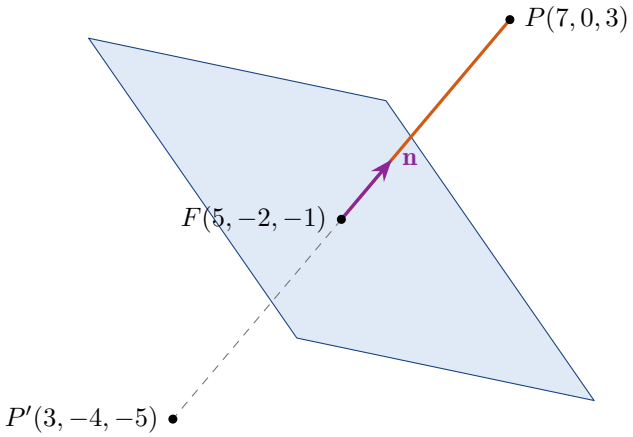
$$\phi = 70.5^\circ \text{ (1 d.p.)}$$

W2.6 Angle between line and plane: sin, not cos



$$\sin\phi = \frac{|\mathbf{b}\cdot\mathbf{n}|}{|\mathbf{b}||\mathbf{n}|}$$

W2.7 Foot of perpendicular and reflection: point and plane

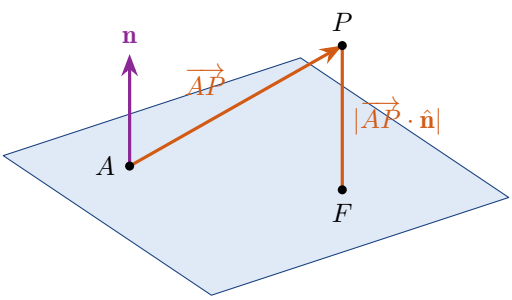


$$l_{PF} : \mathbf{r} = \begin{pmatrix}7\\0\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\2\end{pmatrix},\ \lambda\in\mathbb{R}$$

$$\left(\begin{pmatrix}7\\0\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\2\end{pmatrix}\right)\cdot\begin{pmatrix}1\\1\\2\end{pmatrix} = 1$$

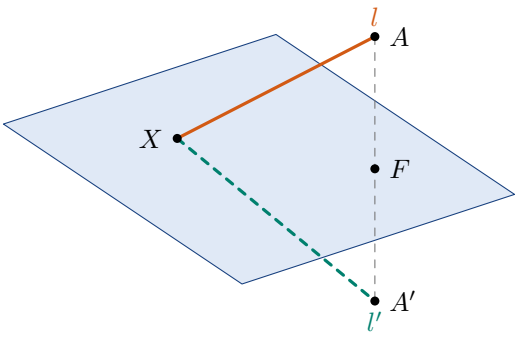
$$\begin{aligned}13 + 6\lambda &= 1\\ \lambda &= -2\\ \therefore F &= (5,-2,-1)\\ \overrightarrow{OF} &= \frac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}\\ \overrightarrow{OP'} &= 2\overrightarrow{OF} - \overrightarrow{OP}\\ &= 2\begin{pmatrix}5\\-2\\-1\end{pmatrix} - \begin{pmatrix}7\\0\\3\end{pmatrix}\\ &= \begin{pmatrix}3\\-4\\-5\end{pmatrix}\end{aligned}$$

W2.8 Distance from point to plane



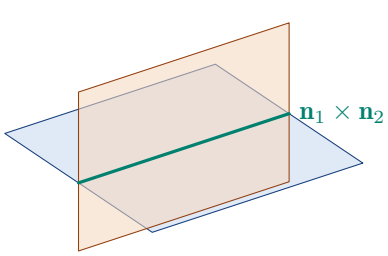
$$
\begin{aligned}
PF &= |\overrightarrow{AP} \cdot \hat{\mathbf{n}}| \\
A(1,0,0) \in p,\ \overrightarrow{AP} &= \begin{pmatrix} 6 \\ 0 \\ 3 \end{pmatrix} \\
PF &= \frac{\left| \begin{pmatrix} 6 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right|}{\sqrt{6}} \\
&= \frac{12}{\sqrt{6}} \\
&= 2\sqrt{6}
\end{aligned}
$$

W2.9 Reflection of a line in a plane

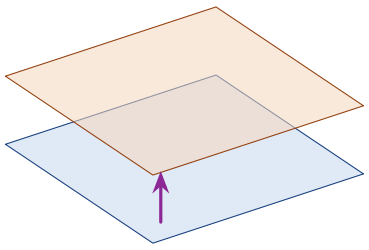


$$
\begin{aligned}
&X = l \cap p \\
&A \in l \Rightarrow F \Rightarrow \overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA} \\
&l' : \mathbf{r} = \overrightarrow{OX} + \mu \overrightarrow{XA'} \\
&\mathbf{b} \cdot \mathbf{n} = 0\ (l \parallel p) :\ \ l' : \mathbf{r} = \overrightarrow{OA'} + \mu \mathbf{b}
\end{aligned}
$$

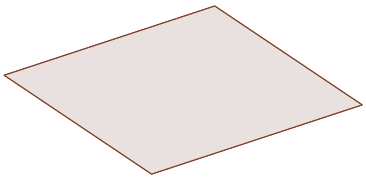
W2.10 Plane–plane: intersecting, parallel, same plane



intersect in a line



$\mathbf{n}_1 \parallel \mathbf{n}_2$, parallel

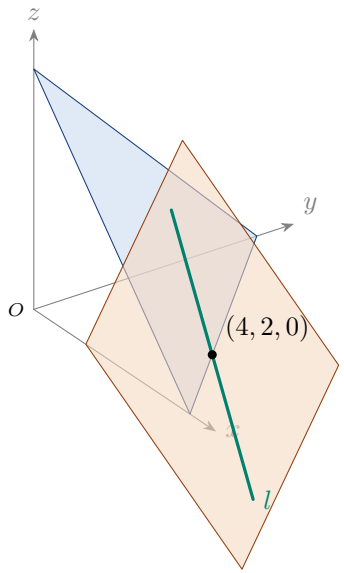


same plane

$$
\begin{aligned}
\mathbf{n}_1 \neq k\mathbf{n}_2 &\Rightarrow \text{line, direction } \mathbf{n}_1 \times \mathbf{n}_2 \\
\mathbf{n}_1 = k\mathbf{n}_2,\ d_1 \neq kd_2 &\Rightarrow \text{parallel} \\
\mathbf{n}_1 = k\mathbf{n}_2,\ d_1 = kd_2 &\Rightarrow \text{same plane}
\end{aligned}
$$

Example 13.5.22

$p_1 : x + y + z = 6$, $p_2 : x - y + 2z = 2$. Find the line of intersection and the acute angle between p_1 and p_2.

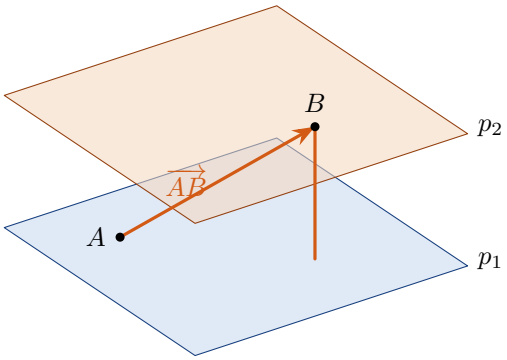


Solution.

$$\begin{aligned}
\mathbf{n}_1 \times \mathbf{n}_2 &= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \\
&= \begin{pmatrix} 1(2) - 1(-1) \\ -(1(2) - 1(1)) \\ 1(-1) - 1(1) \end{pmatrix} \\
&= \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
&\text{Let } z = 0 \\
&x + y = 6 \\
&x - y = 2 \\
&\therefore x = 4,\ y = 2 \\
&l : \mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix},\ \lambda \in \mathbb{R} \\
&\cos\theta = \frac{\left| \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \right|}{\sqrt{3}\sqrt{6}} \\
&\quad = \frac{2}{\sqrt{18}} \\
&\theta = 61.9^\circ \text{ (1 d.p.)}
\end{aligned}$$

W2.11 Distance between parallel planes



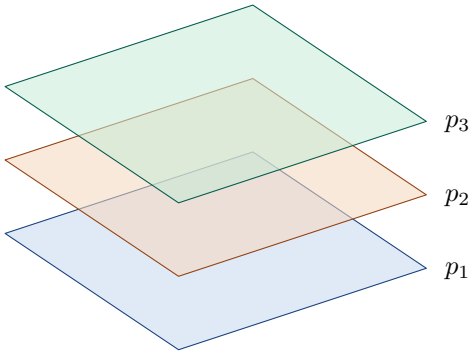
$$\begin{aligned}\text{distance} &= |\overrightarrow{AB} \cdot \hat{\mathbf{n}}| \\ &= \frac{|d_2 - d_1|}{|\mathbf{n}|} \quad (\text{same } \mathbf{n})\end{aligned}$$

$$p_1 : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 1, \; p_2 : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 13 \Rightarrow \frac{|13 - 1|}{\sqrt{6}} = 2\sqrt{6}$$

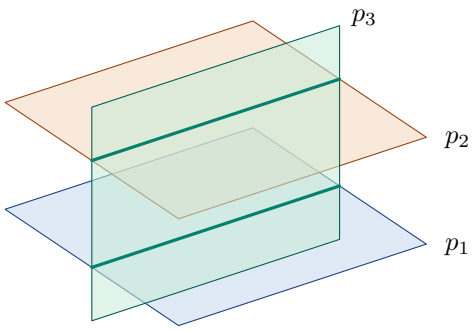
13.6 Interaction between Three Planes

$p_1 : \mathbf{r} \cdot \mathbf{n}_1 = d_1$, $p_2 : \mathbf{r} \cdot \mathbf{n}_2 = d_2$, $p_3 : \mathbf{r} \cdot \mathbf{n}_3 = d_3$ distinct. $l = p_1 \cap p_2$ $(\mathbf{n}_1 \nparallel \mathbf{n}_2)$, $l : \mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

W3.1 All parallel: $\mathbf{n}_1 \parallel \mathbf{n}_2 \parallel \mathbf{n}_3$

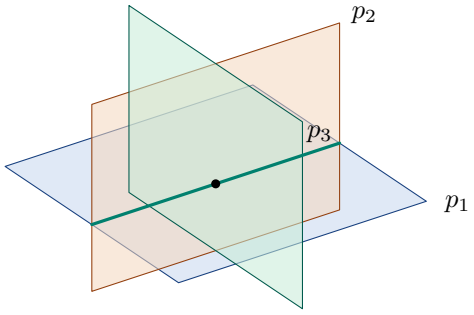


W3.2 Two parallel, one not: $\mathbf{n}_1 \parallel \mathbf{n}_2$, $\mathbf{n}_3 \nparallel \mathbf{n}_1$

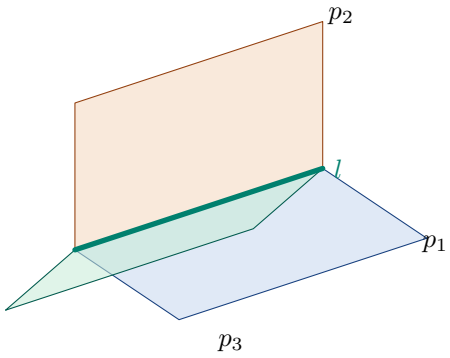


two parallel lines, no common point

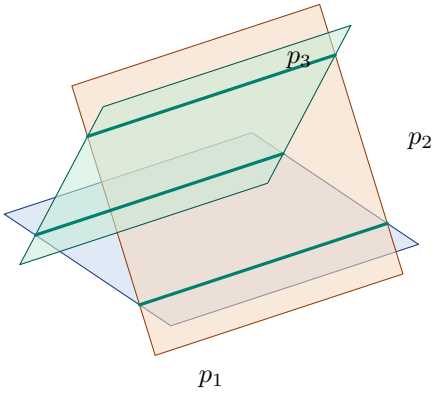
W3.3 One common point: l meets p_3, $\mathbf{b} \cdot \mathbf{n}_3 \neq 0$



W3.4 Common line: $l \subset p_3$, $\mathbf{b} \cdot \mathbf{n}_3 = 0$, $\mathbf{a} \cdot \mathbf{n}_3 = d_3$



W3.5 Triangular prism: $\mathbf{b} \cdot \mathbf{n}_3 = 0$, $\mathbf{a} \cdot \mathbf{n}_3 \neq d_3$



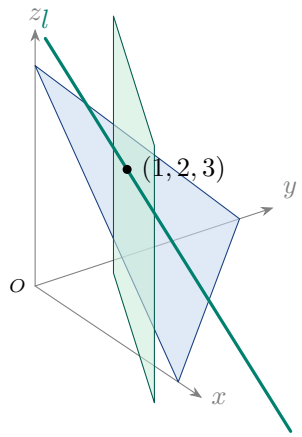
three parallel lines, no common point

W3.6 Summary

Case	Condition	GC (PlySmlt2)
1. all parallel	$\mathbf{n}_1 \parallel \mathbf{n}_2 \parallel \mathbf{n}_3$	no solution
2. two parallel, one not	$\mathbf{n}_1 \parallel \mathbf{n}_2$, $\mathbf{n}_3 \nparallel \mathbf{n}_1$	no solution
3. one point	$\mathbf{b} \cdot \mathbf{n}_3 \neq 0$	unique solution
4. common line	$\mathbf{b} \cdot \mathbf{n}_3 = 0$, $\mathbf{a} \cdot \mathbf{n}_3 = d_3$	infinitely many, 1 parameter
5. triangular prism	$\mathbf{b} \cdot \mathbf{n}_3 = 0$, $\mathbf{a} \cdot \mathbf{n}_3 \neq d_3$	no solution

Example 13.6.1

$p_1 : x + y + z = 6$, $p_2 : 2x - y + z = 3$, $l = p_1 \cap p_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$. Describe the intersection of p_1, p_2, p_3 for (a) $p_3 : x + 2y = 5$, (b) $p_3 : x - 2y = -3$.



Solution.

(a) $$\begin{pmatrix} 3+2t \\ 3+t \\ -3t \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = 5$$

$$9 + 4t = 5$$

$$t = -1$$

$\therefore$ one point $(1, 2, 3)$

(b) $$\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = 0$$

$$\begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = -3$$

$\therefore l \subset p_3$

$\therefore$ common line l